

# Two-Sector DMP Model with Quality Sorting (Annotated)

Based on Mortensen-Pissarides (1994) and Lagos (2006)

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## 1 Environment

- **Time:** Continuous.
- **Sectors:**  $j \in \{H, L\}$  denoting High-tech and Low-tech (non-high-tech). Sector-specific productivity factors are  $p_H$  and  $p_L$ .
- **Workers:** Unit mass of risk-neutral workers. There are 3 global worker types (abilities) indexed by  $i \in \{1, 2, 3\}$ , where ability scalars are  $a_1 < a_2 < a_3$ .
- **Labor Force:** The total mass of type  $i$  workers is  $m_i$ . Workers discount the future at rate  $r$ . Unemployed workers receive a flow benefit  $b$ .
- **Firms:** Large pool of risk-neutral potential firms. A firm can open a vacancy in sector  $j$  at a flow cost  $c_j$ .

## 2 Search and Matching

Workers direct their search to a specific sector based on expected utility, but match randomly with firms within that sector.

- **Sector Choice:** Let  $\phi_{iH} \in [0, 1]$  be the probability/fraction of unemployed type  $i$  workers who choose to search in sector  $H$ , and  $\phi_{iL} = 1 - \phi_{iH}$  for sector  $L$ .
- **Sector Pools:** The total mass of unemployed workers searching in sector  $j$  is  $u_j = \sum_{i=1}^3 \phi_{ij} u_i$ , where  $u_i$  is the number of unemployed type  $i$  workers.
- **Market Tightness:** Let  $v_j$  be the number of vacancies in sector  $j$ . Market tightness is  $\theta_j = v_j/u_j$ .
- **Contact Rates:** The matching function is  $M(u_j, v_j)$ . The worker job-finding rate is  $\lambda(\theta_j) = M(u_j, v_j)/u_j$ . The firm vacancy-filling rate is  $q(\theta_j) = M(u_j, v_j)/v_j$ .
- **Pool Composition:** Conditional on a firm in sector  $j$  meeting a worker, the probability the worker is of type  $i$  is  $\pi_{ij} = (\phi_{ij} u_i)/u_j$ .

### 3 Production and Endogenous Separation

- **Match Quality:** Upon meeting, the firm and worker draw a match-specific idiosyncratic quality shock  $x \sim G(x)$ .
- **Output:** A producing match between a sector  $j$  firm and a type  $i$  worker with shock  $x$  yields flow output  $y_{ij}(x) = p_j a_i x$ .
- **Shocks:** New idiosyncratic shocks  $x'$  arrive at Poisson rate  $\delta$ , drawn from the same distribution  $G(\cdot)$ .
- **Exogenous Separation:** Matches are destroyed for exogenous reasons at rate  $s$ .
- **Endogenous Separation:** A match is formed (and continued) if and only if the joint surplus is non-negative, which defines a reservation match quality  $R_{ij}$ . If  $x < R_{ij}$ , the match separates (or is not formed).

### 4 Value Functions

#### 4.1 Workers

The value of unemployment for a worker of type  $i$  searching in sector  $j$  is  $U_{ij}$ :

$$\underbrace{rU_{ij}}_{\text{Flow value of unemployment}} = \underbrace{b}_{\text{Flow benefit}} + \underbrace{\lambda(\theta_j)}_{\text{Job finding rate}} \underbrace{\int_{R_{ij}}^{\infty} [W_{ij}(x) - U_{ij}] dG(x)}_{\text{Expected gain from finding a job}} \quad (1)$$

The value of employment  $W_{ij}(x)$  satisfies:

$$\begin{aligned} \underbrace{rW_{ij}(x)}_{\text{Flow value of employment}} &= \underbrace{w_{ij}(x)}_{\text{Wage}} + \underbrace{\delta \int_{R_{ij}}^{\infty} [W_{ij}(x') - W_{ij}(x)] dG(x')}_{\text{Expected change in value from a new shock } x' \geq R_{ij}} \\ &\quad - \underbrace{(\delta G(R_{ij}) + s)[W_{ij}(x) - U_{ij}]}_{\text{Expected loss from endogenous or exogenous separation}} \end{aligned} \quad (2)$$

**Optimal Sorting:** Workers choose the sector that maximizes their value:

$$\underbrace{U_i}_{\text{Optimal unemployment value}} = \max\left\{ \underbrace{U_{iH}}_{\text{Value in High-tech}}, \underbrace{U_{iL}}_{\text{Value in Low-tech}} \right\} \quad (3)$$

- If  $U_{iH} > U_{iL}$ , then  $\phi_{iH} = 1$ .
- If  $U_{iH} < U_{iL}$ , then  $\phi_{iH} = 0$ .
- If  $U_{iH} = U_{iL}$ ,  $\phi_{iH} \in [0, 1]$  (indifferent).

## 4.2 Firms

The value of a vacancy in sector  $j$  is  $V_j$ :

$$\underbrace{rV_j}_{\text{Flow value of vacancy}} = \underbrace{-c_j}_{\text{Flow cost of vacancy}} + \underbrace{q(\theta_j)}_{\text{Vacancy filling rate}} \underbrace{\sum_{i=1}^3 \pi_{ij}}_{\text{Expected over worker types}} \underbrace{\int_{R_{ij}}^{\infty} [J_{ij}(x) - V_j] dG(x)}_{\text{Expected firm gain from a match}} \quad (4)$$

Free entry implies  $V_j = 0$  for all active sectors.

The value of a filled job  $J_{ij}(x)$  satisfies:

$$\begin{aligned} \underbrace{rJ_{ij}(x)}_{\text{Flow value of filled job}} &= \underbrace{p_j a_i x}_{\text{Match output}} - \underbrace{w_{ij}(x)}_{\text{Wage cost}} + \underbrace{\delta \int_{R_{ij}}^{\infty} [J_{ij}(x') - J_{ij}(x)] dG(x')}_{\text{Expected change in firm value from a new shock } x' \geq R_{ij}} \\ &- \underbrace{(\delta G(R_{ij}) + s) J_{ij}(x)}_{\text{Expected loss of firm value from separation (using } V_j = 0)} \end{aligned} \quad (5)$$

## 5 Surplus and Equilibrium Conditions

Wages are determined by continuous Nash Bargaining, where workers have bargaining power  $\beta$ :

$$\underbrace{W_{ij}(x) - U_{ij}}_{\text{Worker's share of surplus}} = \underbrace{\beta}_{\text{Worker's bargaining power}} \underbrace{S_{ij}(x)}_{\text{Total match surplus}} \quad (6)$$

$$\underbrace{J_{ij}(x) - V_j}_{\text{Firm's share of surplus}} = \underbrace{(1 - \beta)}_{\text{Firm's bargaining power}} \underbrace{S_{ij}(x)}_{\text{Total match surplus}} \quad (7)$$

The total match surplus  $S_{ij}(x) = W_{ij}(x) - U_{ij} + J_{ij}(x) - V_j$  satisfies:

$$\underbrace{(r + s + \delta) S_{ij}(x)}_{\text{Effective discount rate} \times \text{Total Surplus}} = \underbrace{p_j a_i x}_{\text{Flow output}} - \underbrace{r U_{ij}}_{\text{Worker's outside option}} + \underbrace{\delta \int_{R_{ij}}^{\infty} S_{ij}(x') dG(x')}_{\text{Expected future surplus from new shocks}} \quad (8)$$

### 5.1 Job Destruction (Reservation Quality)

The reservation quality  $R_{ij}$  is defined by  $S_{ij}(R_{ij}) = 0$ :

$$\underbrace{p_j a_i R_{ij}}_{\text{Output at reservation quality}} - \underbrace{r U_{ij}}_{\text{Outside option}} + \underbrace{\frac{\delta}{r + s + \delta} \int_{R_{ij}}^{\infty} p_j a_i (x' - R_{ij}) dG(x')}_{\text{Option value of retaining a marginally viable match}} = 0 \quad (9)$$

### 5.2 Job Creation (Free Entry)

Imposing  $V_j = 0$  yields the job creation condition for sector  $j$ :

$$\underbrace{\frac{c_j}{q(\theta_j)}}_{\text{Expected recruitment cost}} = \underbrace{(1 - \beta)}_{\text{Firm's surplus share}} \underbrace{\sum_{i=1}^3 \pi_{ij}}_{\text{Sum over worker types}} \underbrace{\int_{R_{ij}}^{\infty} S_{ij}(x) dG(x)}_{\text{Expected total surplus upon meeting}} \quad (10)$$

### 5.3 Steady-State Flow Equations

To close the model, one formulates the steady-state unemployment equations for each type  $i$ , equating flows into and out of unemployment. Let  $e_i$  be the number of employed type  $i$  workers.

$$\underbrace{u_i}_{\text{Unemployed type } i} + \underbrace{e_i}_{\text{Employed type } i} = \underbrace{m_i}_{\text{Total mass of type } i} \quad (11)$$

The flow into employment must equal the flow out of employment:

$$\underbrace{\lambda(\theta_j)\phi_{ij}u_i(1 - G(R_{ij}))}_{\text{Flow into employment in sector } j} = \underbrace{e_{ij}(\delta G(R_{ij}) + s)}_{\text{Flow out of employment in sector } j} \quad (12)$$

where  $e_{ij}$  is the mass of type  $i$  workers employed in sector  $j$ .

## 6 Verification of Key Propositions (Analytical Sketch)

**Initial Sorting Assumption:** Suppose initially  $p_H > p_L$  such that Type 3 workers strictly prefer H ( $U_{3H} > U_{3L}$ ), Type 2 workers are on the margin or prefer L ( $U_{2H} < U_{2L}$ ), and Type 1 prefer L ( $U_{1H} < U_{1L}$ ).

**Proposition 1: Higher productivity in H attracts lower quality workers.**

As  $p_H$  increases, the surplus  $S_{iH}(x)$  increases for all  $i$ , causing  $U_{iH}$  to rise. Eventually,  $U_{2H}$  exceeds  $U_{2L}$ . Type 2 workers (who are lower quality than the incumbent Type 3 workers) now switch their search to Sector H. The average quality of workers in Sector H drops because it now includes the lower-ability Type 2 cohort.

**Proposition 2: Rising H-tech productivity generates wage gains outside H-tech.**

When Type 2 workers leave Sector L to search in Sector H, the unemployment pool in Sector L ( $u_L$ ) shrinks. The Job Creation condition in L depends on the expected surplus of meeting a worker. Since Type 1 remains in L, the firm composition  $\pi_{1L}$  shifts. Even holding vacancies fixed, a drop in  $u_L$  increases market tightness  $\theta_L = v_L/u_L$ . A higher  $\theta_L$  increases the job finding rate  $\lambda(\theta_L)$  for the remaining Type 1 workers. This increases their outside option  $U_{1L}$ , which, via Nash bargaining ( $w_{ij}(x) = \beta p_j a_i x + (1 - \beta)rU_{ij}$ ), directly drives up the wages for the non-high-tech workers.

**Proposition 3: Entire labor market becomes tighter.**

In Sector H, higher  $p_H$  directly increases expected firm surplus, incentivizing a massive entry of vacancies  $v_H$ . This pushes  $\theta_H$  up. In Sector L, the outflow of Type 2 workers directly reduces the denominator  $u_L$ , pushing  $\theta_L$  up. Consequently, job finding rates rise across both sectors, making the entire economy tighter.